



Ca/Sr and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios as tracers of Ca and Sr cycling in the Rio Icacos watershed, Luquillo Mountains, Puerto Rico

Julie C. Pett-Ridge*, Louis A. Derry, Jenna K. Barrows

Department of Earth and Atmospheric Sciences, Snee Hall, Cornell University, Ithaca, NY 14853 USA

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ABSTRACT

We investigated Ca and Sr cycling in a humid tropical forest by analyzing Ca/Sr ratios and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios in soil minerals, soil exchangeable cations, soil porewater, and plant roots, wood and leaves, and calculating the relative contributions of Sr from atmospheric inputs and weathering of local bedrock. An unexpectedly large contribution of bedrock-derived Sr and presumably Ca is cycled through the vegetation, reflecting the important role of geological processes in controlling the cycling of base cation nutrients even in a system with intensely weathered soil. This is surprising because over 99% of the Ca and Sr that was originally in the bedrock is leached out of the soil and saprolite during early stages of weathering at this site, and because there are large atmospheric inputs to the site of both sea salt and Saharan dust. Substantial differences in Ca and Sr cycling are seen on small spatial scales between a ridgetop and an adjacent steep hillslope site. Measured Ca/Sr ratios reflect fractionation between these elements during biogeochemical cycling. Fractionation was particularly evident between wood and foliar tissue, but fractionation during soil exchange processes is also likely. In comparing the Ca/Sr ratios of plants, exchangeable cations, and bulk soils, we found that foliar Ca/Sr ratios were greater than exchangeable cation Ca/Sr ratios, which in turn were greater than soil Ca/Sr ratios, similar to patterns observed at other highly weathered tropical sites.

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1. Introduction

The supply of essential base cation nutrients (Ca, Mg, and K) to plants depends on input fluxes from both atmospheric deposition and mineral weathering, and on biologic cycling. Plants growing on highly weathered tropical soils are generally expected to be more dependent on atmospheric sources and internal cycling for these essential elements because intense leaching has depleted them from the soil over time (Jordan, 1985). A shift in the dominant source of base cations from bedrock-derived to atmospheric-derived along a chronosequence of soils in Hawaii supports this generalization (Kennedy et al., 1998; Chadwick et al., 1999). A similar transition is seen along a rainfall gradient in Hawaii, where plants on the most highly leached soils are most dependent on atmospheric inputs (Stewart et al., 2001). Two studies from different areas in Costa Rica, however, have shown that unexpectedly large contributions of base cations from weathering can persist in highly weathered ultisols and oxisols due to either high erosion rates or deep plant rooting (Bern et al., 2005; Porder et al., 2006). Simple generalizations about base cation nutrient cycling in tropical systems are difficult because of the range in (among other things) rates of atmospheric deposition, hydrologic conditions, and the age and type of parent material. Additionally, it is difficult to

quantify weathering and internal nutrient recycling, although approaches such as tracer additions (Kennedy et al., 2002; Dasch et al., 2006) and Ca isotope studies (Wiegand et al., 2005) provide means of assessing these rates.

Ca is an essential plant nutrient which plays a key role in cell walls and plasma membranes and plant growth and development processes (Marschner, 1995). Ca availability is believed to limit productivity in some tropical forests (Cuevas and Medina, 1988; Medina and Cuevas, 1989; Silver, 1994). Ca budgets have recently been a focus of ecological studies due to the potential for human disturbance to accelerate Ca leaching from soils with potentially negative effects to the ecosystem. Acid rain has been shown to cause Ca leaching from soils in U.S. and European temperate forests, which may be causing the declining health of certain tree species (Likens et al., 1996; Shortle et al., 1997; Juice et al., 2006). Acid deposition has decreased in these areas recently due to environmental regulation, but in tropical latitudes acid deposition is expected to increase for decades. Tropical soils typically have little buffering capacity and therefore may be sensitive to acid-induced Ca losses, but to date little research has been done assessing the susceptibility of tropical forests to acid deposition. Particularly in the tropics where N may be less limiting, elevated N deposition associated with human activities may lead to enhanced base cation leaching and soil acidification (Matson et al., 1999). In addition to leaching losses, forest harvesting is another threat to soil Ca availability that has been a particular concern in the tropics (Jordan and Herrera, 1981).

* Corresponding author. Current address: Department of Earth Sciences, University of Oxford, Parks Road, Oxford, OX1 3PR UK. Tel.: +44 1865 272124; fax: +44 1865 272072.
E-mail address: juliep@earth.ox.ac.uk (J.C. Pett-Ridge).

Sr is useful as a tracer of ecosystem Ca sources in biogeochemical studies (Miller et al., 1993; Aberg, 1995; Bailey et al., 1996; Capo et al., 1998; Blum et al., 2002). Sr is an alkaline earth element like Ca, and they behave similarly due to their similar hydrated ionic radius ($r_{\text{Ca}} = 0.272 \text{ nm}$, $r_{\text{Sr}} = 0.274 \text{ nm}$) and ionic charge (Marcus and Kertes, 1968). Given a range in $^{87}\text{Sr}/^{86}\text{Sr}$ ratios among Ca sources, Ca/Sr ratios can be used in combination with $^{87}\text{Sr}/^{86}\text{Sr}$ ratios to identify the sources of Ca in plants. $^{87}\text{Sr}/^{86}\text{Sr}$ ratios are ideal source tracers because these isotopes are not fractionated during biogeochemical cycling (and any minor fractionation that may occur is normalized during the measurement procedure).

In order to use Sr isotopes to trace Ca sources and quantify Ca cycling, any fractionation between Ca and Sr during biogeochemical cycling must be quantified. Several studies have noted that Ca/Sr ratios may be fractionated during plant uptake or allocation, and potentially during soil exchange processes. Higher Ca/Sr ratios in foliage than in roots of individual trees have been observed in a variety of temperate forest species (up to as much as 8 times higher in white pine) (Bailey et al., 1996; Poszwa et al., 2000; Watmough and Dillon, 2003; Dasch et al., 2006), indicating that fractionation between the two elements occurs along the transpiration stream. A second possible mechanism for fractionation of Ca/Sr ratios is a slightly stronger affinity of Sr for ion exchange sites than Ca (Bruggenwert and Kamphorst, 1982; Sposito, 1989). The potential for preferential retention of Ca in plants and preferential retention of Sr in soil complicates the application of Ca/Sr ratios and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios to understanding Ca cycling. Better understanding of these fractionations is needed in order to use Sr as a biogeochemical tracer.

The present study investigated Ca and Sr cycling in a small, undisturbed tropical watershed. The Ca/Sr ratios and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios of atmospheric inputs, soil porewater, the exchangeable cation pool, resistate soil, and vegetation samples were analyzed in order to discriminate between atmospheric and weathering sources of base cations in the vegetation, and to evaluate fractionation between Ca and Sr during biogeochemical cycling. We compared

data from three nearby sites, including a ridgetop, a moderate hillslope, and a steep hillslope site. We also evaluated ecosystem Sr fluxes based on the measured Ca/Sr ratios and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios, and the response time over which the system will shift from one nutrient source to another.

2. Site description

The Rio Icacos watershed (Fig. 1) is located in the Luquillo Mountains of eastern Puerto Rico, at between 600 and 800 m elevation. The 326 ha basin is underlain by the Rio Blanco stock, an early Tertiary quartz diorite pluton (Smith et al., 1998). The coarse to medium-grained quartz diorite is dominated by plagioclase and quartz, with lesser amounts of biotite and hornblende, <1% K-feldspar and chlorite, and accessory magnetite, sphene, apatite, and zircon.

The watershed, covered by lower montane wet forest, has a mean annual temperature of 22 °C and a mean annual rainfall of 4235 mm with minor seasonal variation (Larsen and Concepcion, 1998). Earlier studies estimated that cloudwater input contributed 10% of total precipitation in the Luquillo mountains overall (Weaver, 1972). More recently Schellekens et al. (1998) determined that cloudwater inputs at Pico del Este (1015 m elevation and 2 km east of the Rio Icacos basin) were equal to 7% of the total precipitation. For our calculations we estimate that the Rio Icacos basin (which is at lower elevation, partly below the average base of the cloud layer) receives 5% of precipitation as cloudwater input, increasing the total precipitation inputs to 4458 mm yr⁻¹. Average runoff for the catchment is 3839 mm yr⁻¹ (Larsen and Concepcion, 1998). Both the main stem of the Rio Icacos and the tributary Quebrada Guaba are gauged and monitored through the USGS Water, Energy, and Biogeochemical Budgets program.

Regolith in the watershed consists of a ~0.5 to 1.5 m thick bioturbated inceptisol underlain by saprolite which retains the original bedrock texture, ranging in thickness from ~2 m on steep hillslopes to over 8 m on stable ridgetops. The saprolite, in turn, is underlain by partly decomposed 'saprock' underlain by competent quartz diorite.

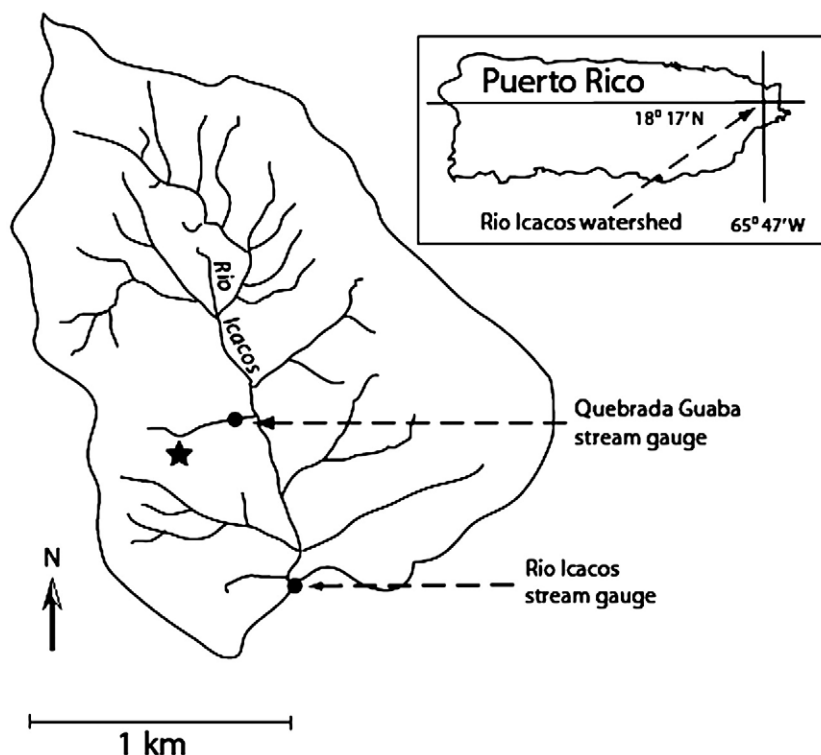


Fig. 1. Map showing Puerto Rico and the Rio Icacos watershed, including the location of the USGS gauging stations on the main stem of the Rio Icacos and on the tributary Quebrada Guaba. The star marks the location of the LG1, LG2, and LG3 sampling sites.

The three studied sites, designated LG1, LG2, and LG3 (White et al., 1998), are situated on the shoulder of the Guaba ridge on the western side of the basin. The LG1 site is located in a relatively flat ridgetop area, minimizing effects of soil creep, erosion, and sediment accumulation. The LG2 site is located on an adjacent steep hillslope (~50% grade), and the LG3 site is located nearby on a moderate hillslope (~25% grade). As is common in tropical forests with high rainfall and tight nutrient cycling, roots in Luquillo Mountain forests are concentrated in the top 30 cm, with very few observed below 70 cm depth (Lawrence, 1996). This was confirmed by our observations of soil pits dug near the study sites in the Rio Icacos watershed.

Solute flux calculations indicate that the Icacos watershed has among the fastest documented granitoid weathering rates in the world (McDowell and Asbury, 1994; White and Blum, 1995). The short-term chemical weathering rate, based on fluxes of major elements in streamwater, translates into a basin-wide mean denudation velocity of between 58 and 75 mm ka⁻¹ (McDowell and Asbury, 1994; Brown et al., 1995; White et al., 1998; Turner et al., 2003). A long-term denudation velocity of 43 ± 15 mm ka⁻¹ was calculated based on ¹⁰Be in fluvial sediments (Brown et al., 1995). The general similarity among the short-term and long-term denudation velocities implies that the system can be considered to be approximately in steady state, and that the regolith thickness remains roughly constant over the timescale of regolith development (Turner et al., 2003). In addition to being in steady state with respect to regolith thickness, the Quaternary uplift rate for northeastern Puerto Rico is estimated to be 40 mm ka⁻¹ (Scatena, 1995), implying that the landscape is roughly in steady state between uplift and erosion over geologic timescales.

3. Methods

We measured Ca/Sr ratios and ⁸⁷Sr/⁸⁶Sr isotope ratios of atmospheric inputs, soil porewater, plant tissue, extracted exchangeable soil cations, acid leachates of the soil, and the remainder of the soil, which we refer to as 'resistate soil'. Open bulk precipitation (rainfall) was sampled using plastic funnels as part of the ongoing USGS Water, Energy, and Biogeochemical Budgets program, and cloudwater was sampled using collectors designed by Scholl et al. (2002). Openfall was collected in a clearing near the Icacos stream gauge, and additional openfall samples as well as cloudwater samples were collected at Pico del Este, which is approximately 2 km to the east of the Rio Icacos basin. Throughfall was collected using 20 cm diameter plastic funnels randomly placed under the closed canopy at 1 m height at each of the three study sites. All precipitation samples were filtered and acidified in the lab, within a few weeks of their collection. An unfiltered aliquot of each cloudwater sample was evaporated down and treated with a concentrated HF–HNO₃–HCl hot plate digestion in order to assess the presence of particulate matter within the samples. Soil porewater was collected using tension lysimeters. Porewater pH was determined immediately after samples were brought to the surface, and porewaters were subsequently filtered at 0.2 µm and acidified. Major ions for water samples were analyzed by ion chromatography and inductively coupled plasma emission spectroscopy, with analytical uncertainty of ~5%. Soil porewater from LG3 was not analyzed.

We collected samples of leaves from each of the dominant plant species at each of the lysimeter field sites. Leaves were pulled directly from standing trees. Stemwood cores were collected from dominant trees at each site. In order to better assess variability between species, leaves were collected from additional species growing nearby but not within the lysimeter field sites. Tree cores and leaves were dried and ground. Roots were picked out of field-moist soils (top 20 cm) from the location of the throughfall collector at each site. Tree fern roots were separated from the LG3 sample, but the species composition of other root samples was not determined. Roots less than 2 mm diameter were washed in de-ionized water, soaked in a 0.6% detergent solution (Tween 20), and rinsed in water again. These roots were then

swirled in a 0.01 M NaEDTA solution, rinsed again, dried and ground. All plant samples were digested in 10 ml of ultra high purity concentrated HNO₃ in a microwave oven.

Soils were collected from hand-augured profiles at LG1 and LG2. Additionally, a 15 cm depth soil sample was collected from all three sites at the location of the throughfall collector, which, at LG1 and LG2, was situated ~2 m away from the location of the hand-augured soils. All soils were dried at 50 °C and passed through a 2 mm sieve. Soil pH was determined on a 1:1 soil-to-water mixture on dried soil. Exchangeable cations were extracted from soils at 5 different depths at each of the LG1 and LG2 sites, as well as the 15 cm depth samples taken from the location of the throughfall collector at each of the sites. For the exchangeable cation extraction, 50 mL of 1 M NH₄OAc was added to 10 g of soil and shaken for 30 min. These were centrifuged and the supernatant was filtered. An additional 50 ml of 1 M NH₄OAc was added to the soils, swirled for 1 minute, then centrifuged, filtered and added to the first supernatant. We investigated the acid-leachable Sr and Ca content of a subset of 5 soil samples (including both shallow and deep samples from both LG1 and LG2) that had already undergone exchangeable cation extraction. The first test used 0.5 M NaOAc adjusted to pH 4.2 with HOAc, and a second test was done on a separate cut of each sample using 1 M HNO₃. In both cases soil samples were swirled occasionally with their extractants over the course of 24 h at room temperature, then centrifuged and filtered. A 0.1 g sample of soil left over from the exchangeable cation extractions, referred to as the "resistate soil", was fully digested in HF–HCl–HNO₃ in a microwave oven for further analysis.

Aliquots of all samples including soil extractions were brought up in 2% HNO₃ for concentration analysis on a Finnigan Element 2 inductively coupled plasma mass spectrometer (ICP-MS). Procedural blanks were collected and monitored for each procedure, and never contributed more than 1% to the Ca and Sr in the samples. Sr separations were performed on remaining sample aliquots using Dowex 50-8X cation exchange resin and Eichrom Sr-Spec resin sequentially. ⁸⁷Sr/⁸⁶Sr ratios were measured on a VG Sector 54 thermal ionization mass spectrometer (TIMS). Total procedural blanks for Sr were negligible. Sr isotopic compositions were corrected for mass fractionation using an exponential law with ⁸⁶Sr/⁸⁸Sr = 0.1194. Analyses of NBS-987 standard yielded an average ⁸⁷Sr/⁸⁶Sr value of

Table 1
Luquillo precipitation samples.

Sample date	Sample ID	Type	Sr ng g ⁻¹	Ca ng g ⁻¹	Ca/Sr molar	⁸⁷ Sr/ ⁸⁶ Sr
17/5/05	LQR3	Rainfall	2.4	578	526	0.71038
14/6/05	LQR3	Rainfall	0.8	208	576	0.71026
14/6/05	LQR4	Rainfall	0.9	203	495	0.71029
6/10/05	LQR3	Rainfall	1.0	265	664	0.71034
17/5/05	LQF2	Cloudwater	7.1	1251	383	0.71047
14/6/05	LQF2	Cloudwater	7.6	1429	414	0.71051
14/6/05	LQF2*	Cloudwater	7.3	1350	406	0.71050
14/7/05	LQF2	Cloudwater	8.0	1586	434	0.71042
14/7/05	LQF2*	Cloudwater	8.6	1727	440	0.71041
24/3/06	LG1	Throughfall	2.1	372	397	0.70993
27/4/06	LG1	Throughfall	2.9	323	246	0.70964
1/5/06	LG1	Throughfall	3.3	443	294	0.70974
15/5/06	LG1	Throughfall	6.9	830	265	0.70957
24/3/06	LG2	Throughfall	2.5	506	437	0.70895
27/4/06	LG2	Throughfall	2.8	593	467	0.70900
1/5/06	LG2	Throughfall	2.5	421	373	0.70898
15/5/06	LG2	Throughfall	4.4	732	365	0.70886
24/3/06	LG3	Throughfall	18.5	1808	214	0.70981
27/4/06	LG3	Throughfall	13.7	1476	236	0.70973
1/5/06	LG3	Throughfall	14.8	1524	226	0.70975
15/5/06	LG3	Throughfall	19.4	1983	224	0.70971

*These samples were not filtered and were fully digested in strong acids.

LQR3 samples were collected near the Icacos stream gauge.

LQR4 and LQF2 samples are from Pico del Este.

Table 2

Bulk soil and saprolite chemistry from LG1 and bedrock chemistry.

Depth cm	Sr $\mu\text{g g}^{-1}$	Ca $\mu\text{g g}^{-1}$	Zr $\mu\text{g g}^{-1}$	Ca/Sr molar	$^{87}\text{Sr}/^{86}\text{Sr}$
15	7	298	214	93	0.71080
46	5				0.71040
61	5	123	123	54	0.71029
76	5				0.70914
122	7				0.70773
152	7	55	130	17	0.70943
183	7				0.70984
244	9	63	127	15	
335	7	217	103	68	0.70826
427	6	77	100	28	
533	49	5350	95	239	
594	97				0.70428
655	143	15,200	96	233	
686	111				0.70421
747	152	16,900	119	243	
Quartz diorite bedrock	301	44,700	92	458	0.70413
Plagioclase (56 wt.%)	545	71,400		286	0.70407
Hornblende (6 wt.%)	77	76,700		2180	0.70583
Biotite (10 wt.%)	19				0.72527

0.71026. The 2σ uncertainties for the isotopic data based on repeat analyses of standards are ± 0.000014 for $^{87}\text{Sr}/^{86}\text{Sr}$.

4. Results

4.1. Atmospheric inputs

Ca and Sr concentrations, Ca/Sr ratios (reported throughout in molar units), and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios for open rainfall and cloudwater samples are presented in Table 1. Average measured concentrations for Sr and Ca are lower in rainfall (1.3 and 313 ng g^{-1} , respectively) than in cloudwater (7.7 and 1469 ng g^{-1}). Average measured Ca/Sr and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios in rainfall are 565 and 0.71032, respectively; in cloudwater they are 415 and 0.71046, respectively. Assuming that 95% of wet precipitation is rainfall and 5% is cloudwater, average Ca/Sr of measured wet precipitation is 558 with an $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of 0.71033 (Table 6).

The two known important components of atmospheric deposition in the Luquillo Mountains that contribute Ca and Sr are sea salt and Saharan dust. Average Ca/Sr and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios of sea salt and Saharan dust are shown in Table 6. The Ca/Sr ratio of sea salt is 115, with an $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of 0.70917 (Dia et al., 1992; Berner and Berner, 1996). Based on available analyses from the literature, Saharan dust transported to the Caribbean has an average value of 195 $\mu\text{g g}^{-1}$ Sr with $^{87}\text{Sr}/^{86}\text{Sr}$ of 0.71788 (Biscaye et al., 1974; Grousset et al., 1988; Grousset et al., 1992; Rognon et al., 1996; Grousset et al., 1998). The average Ca/Sr content of Saharan dust exported over the tropical

North Atlantic is 589 (Moreno et al., 2006). This Ca/Sr ratio is higher than the crustal average ratio because of the significant contribution of calcite and dolomite from the Western Sahara source regions (Schutz and Seibert, 1987; Moreno et al., 2006; Kandler et al., 2007).

The Ca/Sr ratios in our measured rainfall samples do not fall on a simple mixing line between average sea salt and average Saharan dust end-members. This may be related to natural heterogeneity between individual weather events and possible additional sources that contribute Ca and Sr. These other components have not been quantified, but are believed to be less important. Unknown additional atmospheric inputs may include local anthropogenic carbonate dust, however, no significant sources are known upwind (north and east) of the Icacos watershed (Brown et al., 1983). Volcanic ash inputs to Puerto Rican soils are minimal relative to Saharan dust inputs (Muhs et al., 1990; Heartsill-Scalley et al., 2007). We acknowledge the uncertainty regarding atmospheric inputs, but in order to choose a value for modeling purposes we calculate the average based on sea salt and Saharan dust only.

The average $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of the total atmospheric input (defined as the sum of all openfall wet and dry deposition) based on a watershed-scale Sr isotope mass balance is 0.71343, with a total atmospheric Sr input of 1.12 $\text{mol ha}^{-1} \text{yr}^{-1}$ (Pett-Ridge et al., 2009). Using end-member mixing, and the known $^{87}\text{Sr}/^{86}\text{Sr}$ of Saharan dust and sea salt, 49% of the total Sr flux is derived from dust and 51% from sea salt. Using the Ca/Sr ratios of the two end-members, the Ca/Sr ratio of total atmospheric input is 347. Combining this ratio with the total atmospheric Sr input, the total atmospheric Ca input is 392 $\text{mol ha}^{-1} \text{yr}^{-1}$. This indicates that 83% of atmospheric Ca is from dust and 17% is from sea salt. At the nearby El Verde site, Stallard (2001) found that Ca inputs from desert dust equaled 63% of total atmospheric Ca inputs based on National Atmospheric Deposition Program data. We believe that the lower value in the El Verde study is attributable to the fact that only dissolved Ca inputs were included and because rainfall collectors are unlikely to collect dust as efficiently as the forest canopy at the watershed scale.

4.2. Soil samples

Rio Icacos soils are comprised of kaolinite, quartz, goethite, and altered biotite and are not known to contain any primary minerals containing significant Ca or Sr (Pett-Ridge et al., 2009). While the deepest extent of plant roots observed in soil pits was 70 cm, for quantification purposes we have defined the soil conservatively as extending to 1.5 m depth at LG1 and 1 m depth at LG2, our estimate of the maximum rooting depth. Concentrations of Ca and Sr in soil samples were low relative to bedrock (Tables 2 and 3). Within the soil, Ca concentrations are highest in the shallowest samples. The percent of total soil (resistate + exchangeable) Ca that is exchangeable is also

Table 3

Sequential extractions of soil samples from LG1, LG2, and LG3.

Sample ID	Depth cm	Exchangeable Sr $\mu\text{g g}^{-1}$	Resistate Sr $\mu\text{g g}^{-1}$	Exchangeable Ca $\mu\text{g g}^{-1}$	Resistate Ca $\mu\text{g g}^{-1}$	Exchangeable Ca/Sr molar	Resistate Ca/Sr molar	Exchangeable $^{87}\text{Sr}/^{86}\text{Sr}$	Resistate $^{87}\text{Sr}/^{86}\text{Sr}$	pH
LG1 15 tc	15	1.19	7.79	66	247	121	69	0.70951	0.70960	
LG1 15	15	0.88	1.44	129	139	320	211	0.70935	0.71080	4.71
LG1 46	46	0.14	4.91	11	124	182	55	0.70976	0.71042	4.62
LG1 61	61	0.14	3.90	11	103	174	58	0.70948	0.71032	4.58
LG1 76	76	0.22	0.28	3	13	33	102	0.70959	0.70869	4.48
LG1 122	122	0.33	10.95	7	159	44	32	0.70943	0.70707	4.77
LG2 15 tc	15	1.33	3.25	114	160	187	108	0.70899	0.70729	
LG2 15	15	0.13	0.55	12	51	202	201	0.70931	0.70805	4.27
LG2 30	30	0.05	0.42	6	20	252	105	0.70922	0.70968	4.52
LG246	46	0.03	3.46	3	85	176	54	0.70942	0.71080	4.49
LG2 61	61	0.03	1.45	2	23	164	34	0.70939	0.70950	4.67
LG2 91	91	0.10	1.93	2	46	44	52	0.70904	0.70912	4.90
LG3 15 tc	15	1.28	3.26	66	142	113	95	0.70969	0.71231	

"tc" samples were collected at the location of the throughfall collector at each site, ~2 m away from the profile where the rest of the samples were taken.

highest in shallowest samples. In the LG1 soil profile (Table 3), 48% of total soil Ca is exchangeable at 15 cm, while 4% of total soil Ca is exchangeable at 122 cm.

The absence of soil minerals containing appreciable Sr or Ca in our soil samples was indicated by several lines of evidence. The concentrations of Sr and Ca in soil that could be defined as acid-leachable by either protocol (buffered acetic acid or nitric acid) were very low, averaging 4% and 6% of total soil content for Sr and Ca respectively. Even when the length of time exposed to 1 M HNO₃ was extended to 3 days, the amount of Ca and Sr extracted did not increase, and thus we decided not to continue this procedure on the rest of the samples. Weathering losses of Ca and Sr normalized to the least mobile index element (Zr), where a value of 1 indicates no loss relative to bedrock and a value of 0 represents complete loss, are shown in Fig. 3. Both elements exhibit near complete loss during weathering in

Rio Icacos regolith in all samples <4 m depth. XRD analyses confirm the absence of plagioclase and hornblende (the primary mineral hosts of Ca and Sr in the bedrock) in all samples <4 m depth (Pett-Ridge et al., 2009).

⁸⁷Sr/⁸⁶Sr ratios for exchangeable cations show limited variation with depth in the soil (Table 3 and Fig. 2a), but the ⁸⁷Sr/⁸⁶Sr ratios in exchangeable cations at LG2 are slightly lower than those at LG1. Greater variation exists among ⁸⁷Sr/⁸⁶Sr ratios of the resistate soil samples. At LG1, the resistate soil samples show a trend of decreasing ⁸⁷Sr/⁸⁶Sr ratios with depth, but at LG2 no clear trend is apparent. Sr isotope evidence indicates that both kaolinite and altered biotite are actively dissolving in the soil, while new kaolinite is simultaneously precipitating (Pett-Ridge et al., 2009). The ⁸⁷Sr/⁸⁶Sr ratio of soil kaolinite is 0.713, with a Sr concentration of 5 to 7 μg g⁻¹. Altered biotite in soil has 1 to 2 μg g⁻¹ Sr but an insufficient amount was

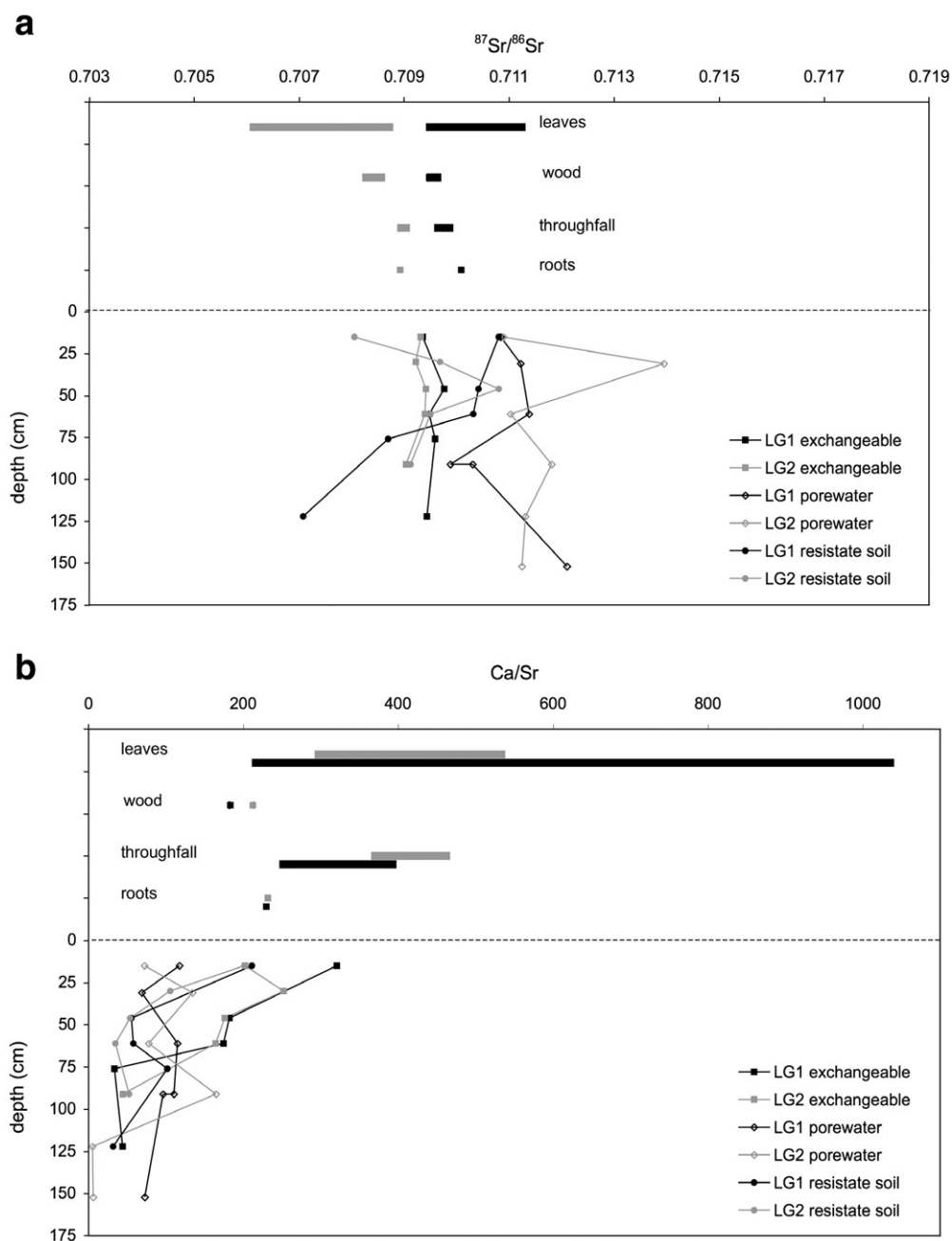


Fig. 2. (a) ⁸⁷Sr/⁸⁶Sr ratios, and (b) Ca/Sr ratios for leaves, wood, throughfall, roots, exchangeable cations, resistate soil, and soil porewaters at the ridgetop (LG1, shown in black) and steep hillslope (LG2, shown in grey) site. The vertical position of data points above the dashed line are arbitrary, but below the dashed line the depth from which the sample was collected is represented.

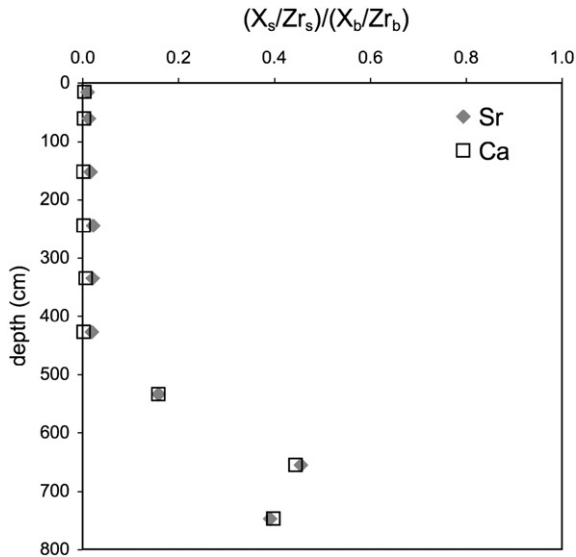


Fig. 3. Losses of Ca and Sr from soil and saprolite at the ridgetop site as indicated by the ratio to the least mobile element Zr, normalized to that ratio in the bedrock. A value of 1 indicates no loss relative to bedrock, and a value of 0 indicates complete loss.

present at 15 cm depth to measure its $^{87}\text{Sr}/^{86}\text{Sr}$ ratio (3% by wt.); at 152 cm depth the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of altered biotite is 0.72892 (Pett-Ridge et al., 2009). The lower $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of the bulk soil (0.708 to 0.711, Table 2) compared to both kaolinite and altered biotite in soil suggests that Sr in organic matter contributes a large portion of the total Sr in the soil. Organic matter, which was included in resistant soil based on our extraction procedure, is likely to have isotopic Sr similar to biomass. Additionally, Sr adsorbed onto and included within Fe-oxides (originally derived from deeper porewaters) may contribute relatively unradiogenic Sr to the soil.

Ca/Sr ratios of exchangeable cations are generally similar between sites (Table 3 and Fig. 2b). Ca/Sr ratios of the resistate soil are generally lower than those of the exchangeable cations at both sites. A depth trend in Ca/Sr ratios of both exchangeable cations and the resistate soil

Table 4
LG1 and LG2 porewater samples.

Sample date	Depth cm	Sr ng g ⁻¹	Ca ng g ⁻¹	Ca/Sr molar	$^{87}\text{Sr}/^{86}\text{Sr}$	pH
11/2/04 LG1 S11	15	2.86	153	117	0.71084	4.44
11/2/04 LG1 S6	31	1.62	51	69	0.71122	4.49
11/2/04 LG1 S4	61	1.19	63	115	0.71138	4.44
11/2/04 LG1 S10	91	5.18	261	110	0.70988	4.58
20/6/03 LG1 S10	91	3.38	149	96	0.71030	
11/2/04 LG1 S7	152	0.98	32	72	0.71210	4.57
20/6/03 LG1 S7	152	0.71	38	115	0.71140	
11/2/04 LG1 S1	181	2.69	143	116	0.71000	4.73
20/6/03 LG1 S1	181	2.32	120	113	0.71009	
11/2/04 LG1 S12	244	1.33	83	137	0.71027	4.54
11/2/04 LG1 S3	366	2.09	244	256	0.71007	4.66
11/2/04 LG1 S14	450	1.79	212	259	0.70963	4.69
11/2/04 LG1 S9	548	1.96	245	273	0.70948	4.60
11/2/04 LG1 S5	671	2.47	220	194	0.70679	5.15
11/2/04 LG1 S13	747	1.54	219	311	0.70937	4.71
11/2/04 LG1 S8	853	3.73	562	330	0.70635	4.59
11/2/04 LG2-S9	15	2.49	82	72	0.71089	4.30
11/2/04 LG2-S8	31	0.45	27	134	0.71395	4.92
11/2/04 LG2-S6	61	0.67	24	78	0.71103	4.90
11/2/04 LG2-S5	91	0.60	45	164	0.71181	4.87
11/2/04 LG2-S7	122	0.82	71	188	0.71131	4.88
11/2/04 LG2-S4	152	0.90	43	104	0.71124	4.85
11/2/04 LG2-S3	183	0.51	40	172	0.71208	4.86
11/2/04 LG2-S2	366	0.79	22	60	0.71041	4.89

is evident, with ratios as high as 320 at the surface and as low as 32 at the base of the soil. In separate bulk soil and saprolite samples from the entire weathering profile at LG1, Ca/Sr ratios exhibit a minimum value of 15 at 244 cm depth (Table 2). Ca/Sr ratios are low throughout the upper saprolite, and higher at saprolite depths of >4 m (where primary bedrock minerals are present in the saprolite) (Fig. 4).

Average soil Ca/Sr and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios for soil exchangeable cations and resistate soil are presented in Table 6. These values are weighted by the mass per unit area of each horizon, determined by the product of Sr and Ca concentrations with horizon thickness and bulk density. The masses of Sr and Ca in the exchangeable cation and resistate soil pools (Table 7) were calculated by multiplying the measured concentration in each horizon by the density and the horizon thickness, and summing over all the soil horizons.

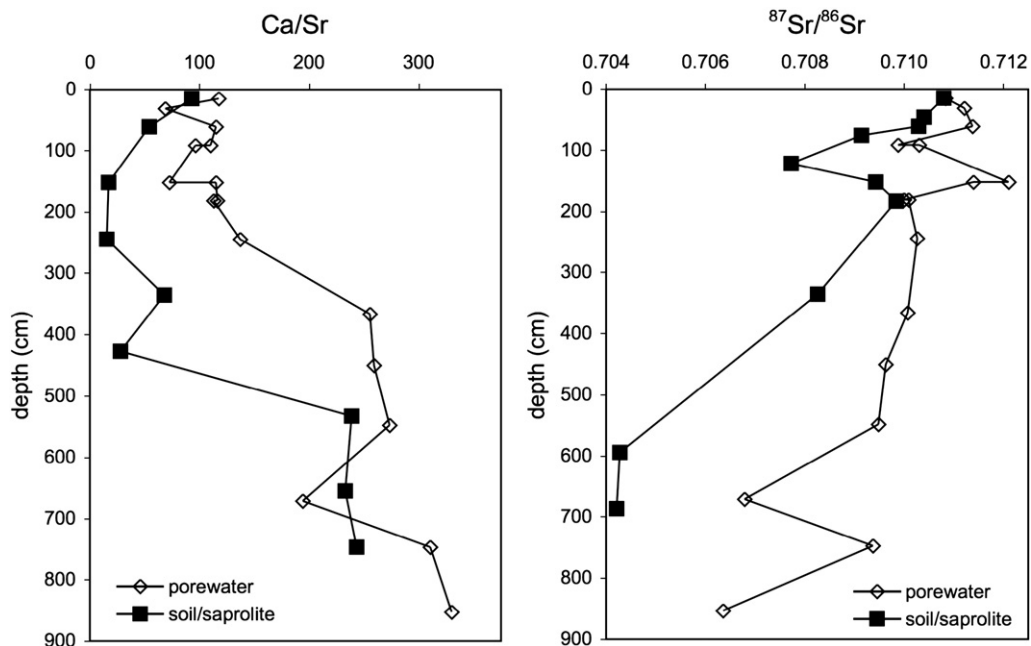


Fig. 4. Ca/Sr ratios and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios for bulk soil and bulk saprolite, and porewaters from the ridgetop site (LG1). The influence of plagioclase weathering is evident at >4 m depth.

Table 5

Plant samples from LG1, LG2, LG3 and surrounding area.

Sample date	Sample ID	Type	Sr $\mu\text{g g}^{-1}$	Ca $\mu\text{g g}^{-1}$	Ca/Sr molar	$^{87}\text{Sr}/^{86}\text{Sr}$
20/6/03	LG1 palo colorado a	Leaves	43	4135	211	0.70947
12/3/06	LG1 palo colorado b	Leaves	26	2762	231	0.70946
20/6/04	LG1 tree fern	Leaves	3	1642	1040	0.71130
20/6/04	LG1 common fern	Leaves	21	2543	260	0.70941
20/6/04	LG2 cecropia a	Leaves	78	10388	292	0.70605
12/3/06	LG2 cecropia b	Leaves	90	13402	326	0.70785
12/3/06	LG2 sierra palm	Leaves	25	6147	538	0.70878
12/3/06	LG3 maricao	Leaves	38	4078	233	0.70963
12/3/06	LG3 bromeliad	Leaves	27	2533	208	0.70971
12/3/06	LG3 matchwood	Leaves	26	5383	455	0.70950
20/6/03	Fire fern	Leaves	39	1149	65	0.70909
20/6/03	Mela stomacae	Leaves	47	7843	366	0.70971
20/6/03	Bamboo	Leaves	10	2909	633	0.70916
20/6/03	Saw grass	Leaves	19	1718	193	0.70959
12/3/06	LG1 roots	Roots	15	1559	230	0.71008
12/3/06	LG2 roots	Roots	11	1216	231	0.70892
12/3/06	LG3 other roots	Roots	32	2013	137	0.70972
12/3/06	LG3 tree fern roots	Roots	14	648	98	0.70978
12/3/06	LG3 bromeliad roots	Roots	126	27597	477	0.70969
12/3/06	LG1 maricao wood	Wood	28	2382	183	0.70943
12/3/06	LG1 palo colorado wood	Wood	6	468	181	0.70970
12/3/06	LG2 cecropia wood	Wood	12	1210	213	0.70820
12/3/06	LG2 sierra palm wood	Wood	35	3398	212	0.70863
12/3/06	LG3 maricao wood	Wood	11	643	125	0.70983
20/6/03	bamboo wood	Wood	1	384	619	0.70904

"a" and "b" samples represent different individuals at the same site. Scientific names are given in the text.

4.3. Porewaters

Ca and Sr concentrations, Ca/Sr ratios, and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios in porewaters are shown in Table 4 and Figs. 2 and 4. A small subset of

Table 6

Average values for each sample type.

Site	Sample type	Ca/Sr molar	$^{87}\text{Sr}/^{86}\text{Sr}$
LG1	Leaves	350	0.70952
LG1	Wood	183	0.70948
LG1	Roots	230	0.71008
LG1	Total biomass	201	0.70960
LG1	Throughfall	293	0.70967
LG1	Soil porewater	104	0.71056
LG1	Exchangeable cations	162	0.70944
LG1	Resistate soil	72	0.70791
LG2	Leaves	358	0.70724
LG2	Wood	212	0.70852
LG2	Roots	231	0.70892
LG2	Total biomass	230	0.70848
LG2	Throughfall	410	0.70893
LG2	Soil porewater	106	0.71137
LG2	Exchangeable cations	192	0.70922
LG2	Resistate soil	86	0.70978
LG3	Leaves	327	0.70962
LG3	Wood	125	0.70983
LG3	Roots	128	0.70974
LG3	Total biomass	143	0.70976
LG3	Throughfall	224	0.70975
LG3	Exchangeable cations	113	0.70969
LG3	Resistate soil	95	0.71231
	Total wet precipitation (measured)	558	0.71033
	Sea salt	115 ^a	0.70917 ^b
	Saharan dust	589	0.71788
	Total atmospheric input (calculated)	347	0.71343
	Bedrock	458	0.70413

Ca/Sr and $^{87}\text{Sr}/^{86}\text{Sr}$ values for LG1 and LG2 reservoirs are weighted averages of all measured samples, weighted by Ca and Sr concentrations, respectively.

^a Sea salt Ca/Sr ratio is taken from Berner and Berner (1996).

^b Sea salt $^{87}\text{Sr}/^{86}\text{Sr}$ is taken from Dia et al (1992).

Table 7

Ca and Sr content of ecosystem reservoirs.

Site	Reservoir	Ca mol ha ⁻¹	Sr mol ha ⁻¹
LG1	Soil porewater	17.68	0.17
LG1	Exchangeable cations	11692	72
LG1	Biomass	8377	42
LG1	Resistate soil	87192	1211
LG1	Total actively cycled pool	20087	114
LG2	Soil porewater	4.45	0.04
LG2	Exchangeable cations	2571	13
LG2	Biomass	12735	55
LG2	Resistate soil	26642	310
LG2	Total actively cycled pool	15311	69

analyses of porewater samples collected at a separate time indicated limited temporal variation in Ca/Sr ratios and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios of soil porewaters. For this analysis we assume that the measured ratios are representative of longer-term averages. The trend in ratios in LG1 saprolite >4 m depth (Fig. 4) can be attributed to dissolution of small amounts of plagioclase present there, which has a higher Ca/Sr ratio and lower $^{87}\text{Sr}/^{86}\text{Sr}$ ratio than the remainder of the soil (Pett-Ridge et al., 2009). Overall, porewater Ca/Sr and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios are intermediate between those of the soil or saprolite at the same depth and those of the atmospheric input values. $^{87}\text{Sr}/^{86}\text{Sr}$ ratios of soil porewaters may be affected by leaching of altered biotite, kaolinite, Saharan dust, or soil organic matter. Relative contributions from individual sources cannot clearly be resolved. However, the small amount of Ca and Sr leached from soil with 1 M HNO₃ treatment suggests that the reservoir of readily dissolvable Ca and Sr in the soils we sampled is limited.

The soil porewater Ca/Sr ratios and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios shown in Fig. 2 also indicate that soil porewater is not in equilibrium with soil exchangeable cations, and additionally that soil porewater does not match throughfall inputs. At both LG1 and LG2 soil porewater has higher $^{87}\text{Sr}/^{86}\text{Sr}$ ratios than both throughfall and exchangeable cations from the same depth (Fig. 2a). Ca/Sr ratios in soil porewaters are lower than both throughfall and exchangeable cations near the surface. Deeper in the soil, soil porewaters have higher Ca/Sr than exchangeable cations at the same depth (Fig. 2b). Site average soil porewater Ca/Sr and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios for LG1 and LG2, weighted for concentration, are presented in Table 6. The Ca/Sr ratios of soil porewater at LG1 and LG2 are 104 and 106, as compared to average throughfall at those sites, 293 and 410. The $^{87}\text{Sr}/^{86}\text{Sr}$ ratios of soil porewater at LG1 and LG2 are 0.71056 and 0.71137, as compared to average throughfall at those sites, 0.70967 and 0.70893. The mass of Ca and Sr in soil porewater (mol ha⁻¹) shown in Table 7 was calculated by multiplying the average concentration of each solute in soil porewater by the depth of the soil (1.5 m at LG1 and 1 m at LG2) and the average soil water content determined by White et al. (1998).

4.4. Vegetation

A range in $^{87}\text{Sr}/^{86}\text{Sr}$ ratios exists between Rio Icacos plant species, but different tissues within individual plants have similar $^{87}\text{Sr}/^{86}\text{Sr}$ ratios (Table 5 and Fig. 2a). Overall, vegetation samples at the steep hillslope site (LG2) have lower $^{87}\text{Sr}/^{86}\text{Sr}$ ratios than vegetation at the other two sites (Table 5 and Fig. 2a). A large range in Ca/Sr ratios was evident in plant tissue samples relative to soil porewater and soil exchangeable cations (Fig. 2b). The lowest foliar Ca/Sr ratio measured was 65 in foliage of a ground fern (family *Gleicheniaceae*), a pioneer species growing on a recent landslide scar approximately 30 m away from the lysimeter plots. The highest foliar Ca/Sr ratio, 1040, was measured in tree fern foliage (*Cyathea arborea*) at the ridgetop site (LG1). Variation in Ca/Sr ratios was also seen between wood and leaves in individual species, in addition to variation between landscape positions. For example, the wood and leaves of an individual sierra palm tree (*Prestoea montana*) had Ca/Sr ratios of

212 and 538, respectively, and the wood and leaves of an individual palo colorado tree (*Cyrilla racemiflora*) had Ca/Sr ratios of 181 and 231, respectively (Table 5).

Table 6 shows weighted average total biomass Ca/Sr and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios, based on the measured Sr and Ca concentrations of roots, wood and leaves and on the amount of each type of plant tissue (g m^{-2}) in Rio Icosos vegetation (Weaver and Murphy, 1990). Soil organic matter was not separately analyzed in this study, but we assume that the low $^{87}\text{Sr}/^{86}\text{Sr}$ ratios of the 15 cm depth resistate and exchangeable soil samples from LG2 (Table 3) are likely due to higher organic matter content (having an $^{87}\text{Sr}/^{86}\text{Sr}$ ratio similar to biomass) at the surface of the soil profile. The biomass Sr and Ca contents (Table 7) were calculated by multiplying the average concentration in each type of material (leaves, wood, and roots) by the abundance of each type (g m^{-2}) in the Rio Icosos watershed (Weaver and Murphy, 1990).

4.5. Throughfall

Ca and Sr concentrations, Ca/Sr ratios, and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios for throughfall samples collected over a 2-month period are shown in Table 1 and Fig. 2. Weighted average throughfall Ca/Sr and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios for each site are 293 and 0.70967 for the ridgetop site (LG1) and 410 and 0.70893 for the hillslope site (LG2) (Table 6). Overall, throughfall had lower Ca/Sr ratios and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios as compared to rainfall (Table 1), and generally had higher Ca and Sr concentrations than rainfall, although direct comparison is limited by the fact that samples were not collected simultaneously. Throughfall at the steep hillslope site (LG2) tended towards higher Ca/Sr ratios and lower $^{87}\text{Sr}/^{86}\text{Sr}$ ratios as compared to the other two sites.

The increased concentrations between rainfall and throughfall and the shifts in both Ca/Sr and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios (Fig. 2) are consistent with throughfall being a mixture of atmospheric and foliar sources. Leaching of dust residing on canopy leaves may contribute a third component to throughfall as has been seen in some previous studies, (Graustein and Armstrong, 1983; Miller et al., 1993), but this would shift the throughfall $^{87}\text{Sr}/^{86}\text{Sr}$ ratio in the opposite direction to what is observed (Fig. 2a). Thus we assume foliar leaching was the dominant component at the time these throughfall samples were collected. The lack of Saharan dust evident in our sampling was not unexpected; the majority of Saharan dust is deposited in just a handful of large dust events per year, and the high frequency of rainfall in the area frequently rinses canopy foliage.

5. Discussion

5.1. Weathering inputs

A simple approach to determining the weathering flux of Sr in the soil is an isotopic mass balance between atmospheric and weathering inputs to actively cycled soil and plant Sr reservoir (Eq. (1)). Weathering fluxes within the soil are difficult to estimate by other means because soil porewater $^{87}\text{Sr}/^{86}\text{Sr}$ ratios cannot be calculated as a mixture of the bulk resistate soil and the throughfall end-members and do not show depth trends within the soil.

$$\left(\frac{^{87}\text{Sr}}{^{86}\text{Sr}}\right)_{\text{active}} = \frac{F_{\text{atmospheric}} \times \left(\frac{^{87}\text{Sr}}{^{86}\text{Sr}}\right)_{\text{atmospheric}} + F_{\text{weathering}} \times \left(\frac{^{87}\text{Sr}}{^{86}\text{Sr}}\right)_{\text{weathering}}}{(F_{\text{atmospheric}} + F_{\text{weathering}})} \quad (1)$$

We assume the actively cycled Sr pool is stable over time because there is no evidence of recent human or natural disturbance at this site. At LG1 the combined biomass, exchangeable cation, and porewater pool Sr pools hold 114 mol ha^{-1} (Table 7). Weighted for the mass of Sr in each component, this combined actively cycled pool has an $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of 0.70950. The Sr atmospheric flux ($F_{\text{atmospheric}}$) is $1.12 \text{ mol ha}^{-1} \text{ yr}^{-1}$ with an $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of 0.71343.

The choice of a weathering end-member is not necessarily straightforward in this highly weathered soil where primary mineral sources of base cations appear to be absent. Given the thick saprolite and the depletion of Ca and Sr at <4 m depth at LG1 (Fig. 3), the use of resistate soil as the weathering end-member seems most appropriate. Using the resistate soil ($^{87}\text{Sr}/^{86}\text{Sr}$ ratio of 0.70791) as the weathering end-member, the weathering flux ($F_{\text{weathering}}$) solved for in Eq. (1) is $2.78 \text{ mol ha}^{-1} \text{ yr}^{-1}$. Given that the measured Sr content of the resistate soil at the ridgetop site is 1210 mol ha^{-1} , a weathering flux of $2.78 \text{ mol ha}^{-1} \text{ yr}^{-1}$ would completely deplete the soil Sr content in 435 years. Considering that the soil residence time on the ridgetop is between 30,000 to 60,000 years, based on the soil thickness and ^{10}Be denudation rates (Brown et al., 1995), this weathering flux appears to be unreasonably large.

The resistate soil analyzed at LG2 cannot be the weathering end-member at that site because the actively cycled Sr pool there has a lower $^{87}\text{Sr}/^{86}\text{Sr}$ (0.70862) than both the resistate soil (0.70978) and the atmospheric input (0.71343). This indicates that the soils we collected at LG2 were not representative of the weathering end-member at that site. An additional complication with using resistate soil as the weathering end-member is that much of the Sr in the resistate soil is held in secondary phases that originally derived Sr in part from atmospheric sources via soil porewater.

We chose instead to use the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of bulk bedrock for the weathering end-member (0.70413), which yields a Sr weathering flux of $0.82 \text{ mol ha}^{-1} \text{ yr}^{-1}$ at LG1 and $1.20 \text{ mol ha}^{-1} \text{ yr}^{-1}$ at LG2 using Eq. (1). These fluxes are shown in Fig. 5. The ratio of the actively cycled pool size to the total inputs indicates that the residence time of Sr in the actively cycled pool is 59 years at LG1 and 30 years at LG2. Using bedrock as the weathering end-member suggests that plant roots have access to relatively fresh bedrock and/or partially weathered saprolite. Possible explanations for the unexpected contribution from bedrock include deep rooting and/or subsurface heterogeneity not characterized by our sampling.

Although differences between individual mineral weathering rates can cause the effective weathering end-members in some soils to be very different from that of the bulk bedrock (e.g. Bailey et al., 1996), using the bulk bedrock value as the weathering end-member is a reasonable assumption for the Rio Icosos watershed given the intensity of weathering and nearly equivalent weathering rates of the primary bedrock minerals at the bedrock interface (aside from quartz, which will not affect Ca and Sr) (Buss et al., 2008). A small amount of plagioclase continues to weather in saprolite at >4 m depth (Pett-Ridge et al., 2009), evident in Fig. 4). However, as plagioclase holds 97% of bedrock Sr and 84% of bedrock Ca, the Ca/Sr and $^{87}\text{Sr}/^{86}\text{Sr}$ signatures of plagioclase are fairly similar to those of the bulk bedrock (Table 2). Hornblende is entirely absent from saprolite at this site (Buss et al., 2008; White et al., 1998), and residual biotite in saprolite has lost 90% of its Sr and is shielded within a kaolinite envelope (Pett-Ridge et al., 2009). We assume the Ca content of biotite and altered biotite to be negligible. Trace apatite (0.7 wt.%) and calcite (0.032 wt.%) dissolve at the earliest stages of weathering and are believed to be absent in both saprolite and underlying saprock (Bullen et al., 1998; Buss et al., 2008).

The probability of deep rooting is low based on the high ratio of annual rainfall to potential evapotranspiration (3.7) in the Rio Icosos watershed (Schenk and Jackson, 2005), and other field observations in the area report roots to only 1 m depth (Schulz and White, 1999; Buss et al., 2008), in agreement with our observations. Fine roots in Rio Icosos watershed soils are concentrated in a dense root mat up to ~8 cm thick on the soil surface (USDA, 2002). While it is possible that some roots extend into partially weathered deep saprolite, we believe that the presence of occasional bedrock corestones within the soil is a more likely explanation for the large flux of bedrock-derived Sr to the actively cycled pool.

Some large bedrock corestones (~1–4 m diameter) survive the intense weathering conditions in the Icosos basin (Fletcher et al.,

2006). Corestones result from spheroidal weathering of the underlying quartz diorite pluton, and they are commonly seen in stream channels in the Rio Icacos watershed. While we did not encounter any such obvious sources of unweathered material in multiple soil pits we dug during sampling, the existence of corestones in nearby stream channels suggests that the regolith is likely to be quite heterogeneous. One can easily envision a scenario where a single tree in a stand may have some roots in contact with a corestone, and through litterfall and subsequent repeated nutrient recycling this nutrient resource could be spread to neighboring vegetation. Typical soil and vegetation sampling regimes are likely to miss these few subsurface

nutrient “hotspots”, but the isotopic tracers used here point to their existence.

Interestingly, the handful of other studies that have examined base cation sources in tropical soils have also found an unexpected amount of bedrock-derived Sr in the exchangeable cation pool, similar to what is seen in the Rio Icacos watershed (Kennedy et al., 1998; Poszwa et al., 2002; Bern et al., 2005). In a classic study of Ca cycling at the Walker Branch watershed in Tennessee, a site which has highly weathered ultisols with no remaining primary minerals, the authors invoked the presence of bedrock “pedestals” to explain a source of Ca that was necessary in their mass balance but absent in their sampling (Johnson and Henderson, 1989).

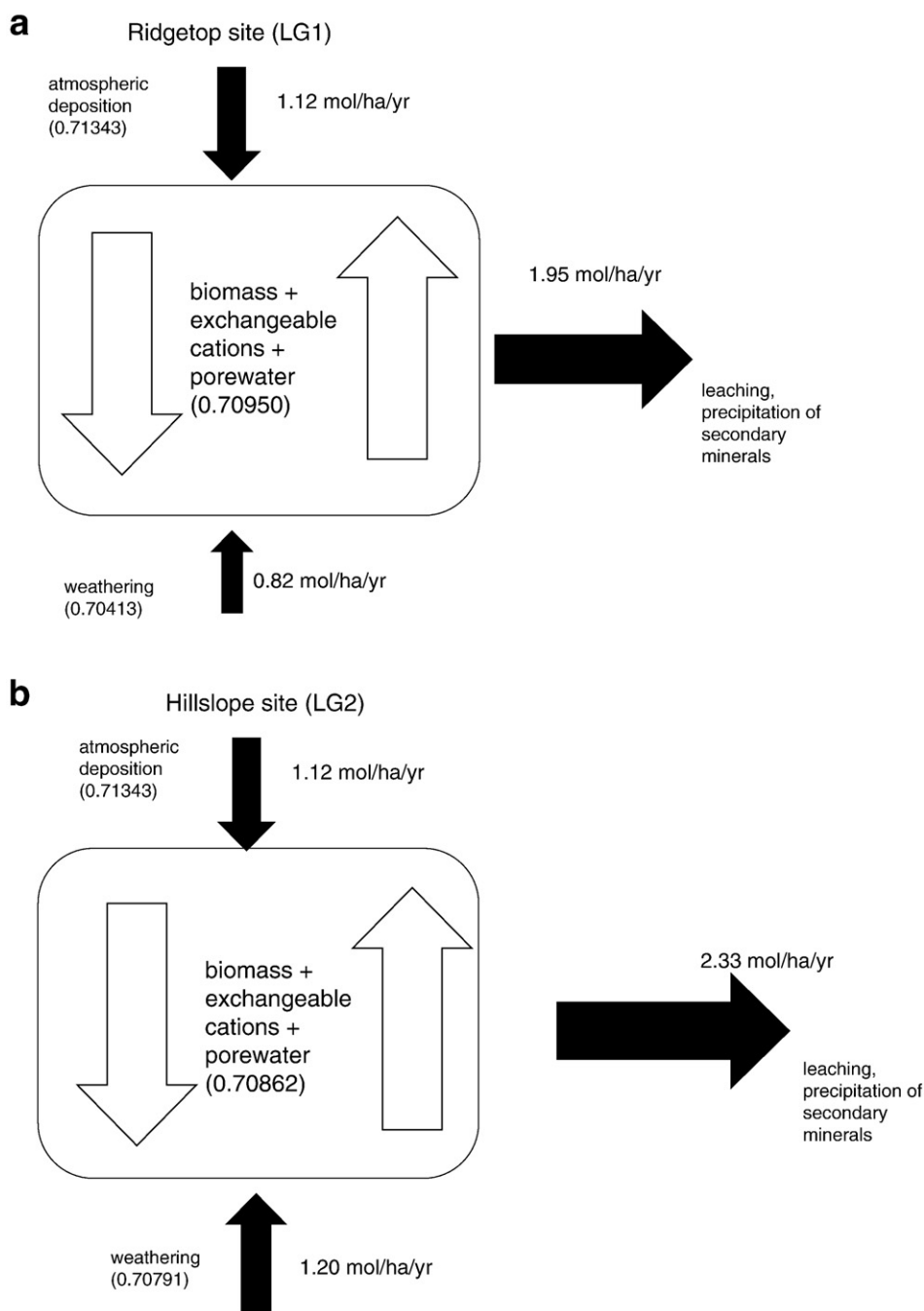


Fig. 5. A simple box model illustrating weathering inputs to the actively cycled Sr reservoir (exchangeable cations + biomass + porewater) based on the Sr isotopic mass balance. Relative sizes of arrows represent relative sizes of fluxes. The fluxes shown do not account for macropore flow (discussed in text). Internal biological cycling fluxes, shown in unfilled arrows, are calculated separately based on the measured Sr content of biomass, and productivity values from (Weaver and Murphy, 1990).

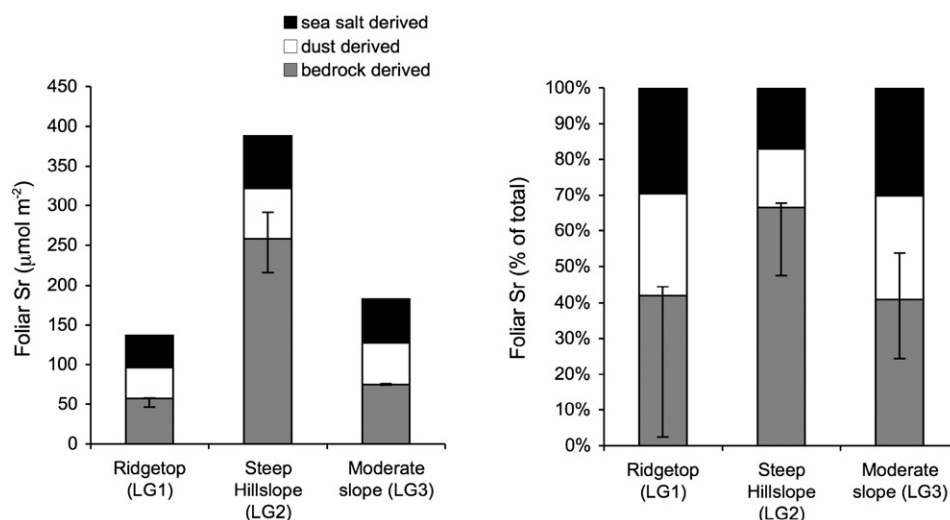


Fig. 6. Sources of foliar Sr for sites LG1, LG2, and LG3, based on weighted average foliar $^{87}\text{Sr}/^{86}\text{Sr}$ values for each site. Sources of foliar Sr are calculated using bedrock ($^{87}\text{Sr}/^{86}\text{Sr}$ of 0.70413) as the weathering end-member. A second mixing calculation was done to divide the total atmospheric end-member ($^{87}\text{Sr}/^{86}\text{Sr}$ of 0.71343) into sea salt and dust components, based on the $^{87}\text{Sr}/^{86}\text{Sr}$ of seawater (0.70917), and the $^{87}\text{Sr}/^{86}\text{Sr}$ of dust (0.71788). Error bars depicted for Sr derived from bedrock show the range of values that are determined if the mixing calculation is done using individual plant $^{87}\text{Sr}/^{86}\text{Sr}$ values instead of a weighted average value for the site.

5.2. Macropore flow

The differences in $^{87}\text{Sr}/^{86}\text{Sr}$ ratios between soil porewater (collected by tension lysimeters) and the exchangeable cation pool (determined through an NH_4OAc extraction) indicates incomplete exchange efficiency between these two pools (Fig. 2 and Table 6). The lack of equilibrium is particularly evident at LG2. Similarly inefficient exchange between these two reservoirs has been observed in other studies (Bullen et al., 1997; Poszwa et al., 2000). Generally speaking, however, geochemical studies assume that these two reservoirs experience rapid and complete equilibration with one another (e.g. Davis and Kent, 1990; Johnson and DePaolo, 1997), and in the majority of studies only the exchangeable cation reservoir is analyzed.

We attribute the lack of equilibrium at these sites to preferential flow in macropores. Hydrological analyses in the Rio Icacos watershed and in the surrounding Luquillo Mountains suggest that macropores are a dominant feature of the soil structure (Larsen and Concepcion, 1998; White et al., 1998; Harden and Scruggs, 2003; Schellekens et al., 2004). We also observed abundant macropores associated with root channels and earthworm activity in Icacos basin soil pits during our sampling.

The inefficient exchange evident between porewater and exchangeable cations suggests that care should be taken when interpreting this type of data and highlights the fact that these sampling methods may target different soil reservoirs. Lysimeters may be sampling soil porewater from macropores rather than porewater that is in equilibrium with the soil exchange complex. The use of zero-tension lysimeters would be expected to exacerbate differences in the observed chemistry between porewater and exchangeable cations.

5.3. Internal recycling and system dynamics

In theory, the signature of bedrock-derived Sr in the actively cycled pool could result from efficient recycling of Sr from earlier stages of weathering when primary minerals were still present, as opposed to being a modern signal. To assess this, we estimated the response timescale over which this system will adjust to changing nutrient sources. Biomass uptake and return fluxes, shown in Fig. 5, were determined using productivity values determined by Weaver and Murphy (1990) and Sr concentrations determined in the current study. Comparison of the total biomass return fluxes to total soil

inputs indicates that internal recycling of Sr accounts for ~55% of Sr inputs to soil. However, the proportion of internal recycling may actually be higher due to macropore flow. If a portion of atmospherically derived Sr bypasses the biomass and exchangeable cation pools in macropores, a smaller weathering flux will be required by the isotopic mass balance than what is shown in Fig. 5. If the input fluxes are smaller, the proportion of internally recycled Sr to newly input Sr in the actively cycled pool increases, as does the residence time of Sr in the actively cycled pool.

While a steady-state assumption for the size of actively cycled Sr pool is reasonable for the Rio Icacos watershed on shorter time-scales, landscape-scale disturbances such as landslides or can periodically perturb nutrient cycling. Landslides in this area occur every 10^4 to 10^5 years on average, exposing less-weathered saprolite and in some cases bedrock (Larsen and Torres-Sanchez, 1998). Landslides deplete the soil's exchangeable base cation reservoir, but at the same time, induce large shifts from atmospheric to bedrock derivation of nutrients, followed by a gradual return towards atmospheric sources as new soils gradually become more weathered.

A simple time-dependent model indicates that the response time of Ca and Sr cycling to such a disturbance in the Rio Icacos watershed will be relatively short. Based on a residence time of actively cycled Sr of 60 years, and assuming 55% internal recycling, it takes just 230 years for the actively cycled pool to make a 90% transition to a new nutrient source. We also considered a scenario in which roughly 50% of inputs bypass the system in macropore flow, which translates to a residence time of ~120 years and internal recycling of ~75%. In this case, the system will undergo 90% transition to a new nutrient source in 960 years. Given the relatively short response timescales as compared to the soil residence times and landslide recurrence interval, and the lack of evidence for recent landslides at the sites we studied, we believe that bedrock-derived Sr in Icacos vegetation likely reflects a current nutrient source, as opposed to efficient recycling of a vestigial source.

These results can be compared to a similar study of Ca cycling in the tropics on the Osa Peninsula in Costa Rica (Bern et al., 2005). In that study the authors reported a large contribution of bedrock weathering to nutrient cycling, which they found surprising. Relative to the Icacos, on the Osa peninsula the size of the actively cycled pool is greater by a factor of 10, and the atmospheric input flux is smaller by a factor of 10. Assuming other variables are similar, the response time

for Ca and Sr cycling to undergo a shift in nutrient sources in the Osa system will be 20 times greater as compared to the Icosos. On the Osa peninsula bedrock Sr isotope signatures could therefore be retained in the vegetation for many thousands of years after the bedrock source is no longer available. Despite this possibility, other evidence from the Osa peninsula, including bedrock fragments in the soil, indicates that the bedrock-derived Sr in Osa vegetation reflects modern sources. Given the very high tectonic uplift rate on the Osa peninsula of 2100 to 6500 mm ka⁻¹ (Bern et al., 2005), the presence of modern bedrock nutrient sources is actually more surprising in the significantly less tectonically active Icosos watershed, with its uplift rate of ~40 mm ka⁻¹ (Scatena, 1995).

5.4. Sources of foliar base cations

The Sr isotope mass balance above indicates that both weathering and atmospheric inputs contribute substantially to actively cycled Sr in the Rio Icosos basin. Sr in leaves may be evaluated as a two-component mixture of atmospheric-derived Sr and weathering-derived Sr in order to directly identify the contribution of each source to vegetation (Eq. (2)). Because the degree of fractionation between Ca and Sr during biological uptake is not known *a priori*, we do not directly extend the mixing calculations to determining the sources of Ca in plants. $X_{\text{weathering}}^{\text{Sr}}$ represents the proportion of foliar Sr that is derived from weathering.

$$X_{\text{weathering}}^{\text{Sr}} = \frac{\left(\frac{{}^{87}\text{Sr}}{{}^{86}\text{Sr}}\right)_{\text{leaves}} - \left(\frac{{}^{87}\text{Sr}}{{}^{86}\text{Sr}}\right)_{\text{atmospheric}}}{\left(\frac{{}^{87}\text{Sr}}{{}^{86}\text{Sr}}\right)_{\text{weathering}} - \left(\frac{{}^{87}\text{Sr}}{{}^{86}\text{Sr}}\right)_{\text{atmospheric}}} \quad (2)$$

The total atmospheric input end-member has an ⁸⁷Sr/⁸⁶Sr ratio of 0.71343, representing the sum of both Saharan dust-derived inputs and sea salt inputs. Using bedrock as the weathering end-member, at LG1, LG2, and LG3, the proportions of foliar Sr derived from bedrock weathering are, respectively, 42%, 67%, and 41% (Fig. 6). Fig. 2 also illustrates that vegetation on the steep hillslope site (LG2) derives a higher fraction of Sr from bedrock than vegetation on the ridgetop and moderate hillslope. Over timescales of tens of thousands of years, hillslopes in this watershed are estimated to have an erosion rate 2.5 to 3.5 times higher than ridgetops (Larsen and Torres-Sanchez, 1998), and this higher erosion rate may contribute to the higher proportion of bedrock Sr in LG2 vegetation.

It is also apparent that the Ca and Sr content of biomass at LG2 is greater than at the other sites (Fig. 6), despite the lower Ca and Sr content of soil at LG2. We do not have sufficient data to assess the relative importance of species effects or topographical effects in accounting for the differences in base cation cycling between LG1 and LG2. In any case, the distribution of species in the Luquillo Mountains is driven in part by topography, with a greater density of fast-growing cecropia (*Cecropia shreberiana*) and sierra palm (*Prestoea montana*) on steep hillslopes (Brown et al., 1983), so it is difficult to separate the two potential drivers.

The fact that the steep hillslope site (LG2) is less than 10 m away from the ridgetop site (LG1) demonstrates that sources of base cation nutrients can be highly variable on a small spatial scale. Similar variability in the proportions of atmospheric and bedrock nutrient sources along topographic gradients has been observed in Hawaii (Porder et al., 2005). The magnitude of the variation in nutrient sources seen within a 10 m radius in Puerto Rico is also comparable to the variation that has been observed across gradients of tens of thousands of years of weathering history (Kennedy et al., 1998) or along sharp rainfall gradients (Stewart et al., 2001) in Hawaii, and suggests large heterogeneity in nutrient availability may be a common feature in montane tropical ecosystems.

The calculation that 42% of foliar Sr is bedrock-derived at the ridgetop site is surprising. The relatively short residence time of the

biomass (28 years, above and belowground (Weaver and Murphy, 1990)), the highly weathered soils, the intensity of leaching, and the large atmospheric inputs to the Icosos basin (located only ~15 km from the coast) are all factors that would suggest very little rock-derived Sr would be found in the biomass. For comparison, the soil residence time on the LG1 ridgetop is between 30,000 to 60,000 years, based on the soil thickness and ¹⁰Be denudation rates (Brown et al., 1995). On the steep hillslope, the soil residence time is estimated to be ~15,000 years, based on the soil thickness and the applying the watershed average denudation rate (65 mm ka⁻¹, (Turner et al., 2003)). As discussed earlier, we believe that the persistence of bedrock corestones in this watershed may account for the presence of bedrock-derived base cations in the vegetation.

5.5. Fractionation between Ca and Sr

The depth gradient in Ca/Sr ratios in soil, shifts between Ca/Sr ratios in throughfall and soil porewater, and shifts between Ca/Sr ratios of different plant tissues (Fig. 2b and Table 6) all suggest fractionation between Ca and Sr during biogeochemical cycling. This fractionation must be characterized in order to use ⁸⁷Sr/⁸⁶Sr ratios as a quantitative tracer of Ca sources and cycling. Fig. 7a and b show mixing diagrams plotting ⁸⁷Sr/⁸⁶Sr ratios versus Ca/Sr ratios of ecosystem components. A handful of samples including porewaters, exchangeable cations, and leaves plot outside the area defined by mixing lines between possible base cation source end-members (bedrock, resistate soil, and atmospheric inputs) as a result of fractionation between Ca and Sr.

Biomass Ca/Sr ratios suggest that fractionation may occur during nutrient uptake from soil and certainly occurs during transport within trees we sampled. Assuming that the pool available for uptake by the biomass is predominantly the exchangeable cation pool, the ratio of the average biomass ratio to the exchangeable cation ratio indicates the discrimination of Ca over Sr during uptake. Using the site-averaged values given in Table 6, at LG1, LG2, and LG3, respectively, this ratio is 1.24, 1.20, and 1.27, indicating preferential uptake of Ca. However, if the majority of biomass uptake is considered to only occur from the surface soil, then no fractionation during uptake is required because Ca/Sr ratios of uppermost exchangeable cation samples are similar to those of the average biomass (Table 3). Fine roots are concentrated in the surface soil in order to take advantage of the nutrient source of litterfall (USDA, 2002; Silver et al., 1994), which suggests that nutrient uptake is predominantly from surface soil.

Results from all three sites (Table 6 and Fig. 2b) indicate fractionation against Sr occurs in plants during transport between wood and foliar tissue, in agreement with previous studies of Ca/Sr fractionation in temperate and sub-alpine forests (e.g. Elias et al., 1982; Bailey et al., 1996; Poszwa et al., 2000; Blum et al., 2000; Watmough and Dillon, 2003; Bullen and Bailey, 2005; Dasch et al., 2006). The differences we observed in Ca/Sr ratios between wood and leaves were much larger than the differences seen between roots and wood, although our lack of paired root and wood samples from individual trees makes it difficult to assess any fractionation that may occur between roots and wood.

The trend of decreasing Ca/Sr ratios in exchangeable cations and in resistate soil in Icosos soils with increasing depth likely reflects a mixture where the contributions to the exchangeable cation pool from decomposing litter are higher at the surface and contributions from resistate weathering are higher at the base of the soil. The soil Ca/Sr ratios could be low at depth through several mechanisms, including possible preferential uptake of Ca by plants and/or preferential release of Ca during weathering.

The large difference between porewater and throughfall Ca/Sr ratios (Fig. 2b and Table 6) shows that fractionation between Ca and Sr occurs when throughfall enters the soil environment. Low concentrations of Sr and Ca in soil porewater relative to throughfall suggest that inputs to soil porewater are limited, although the shift in ⁸⁷Sr/⁸⁶Sr

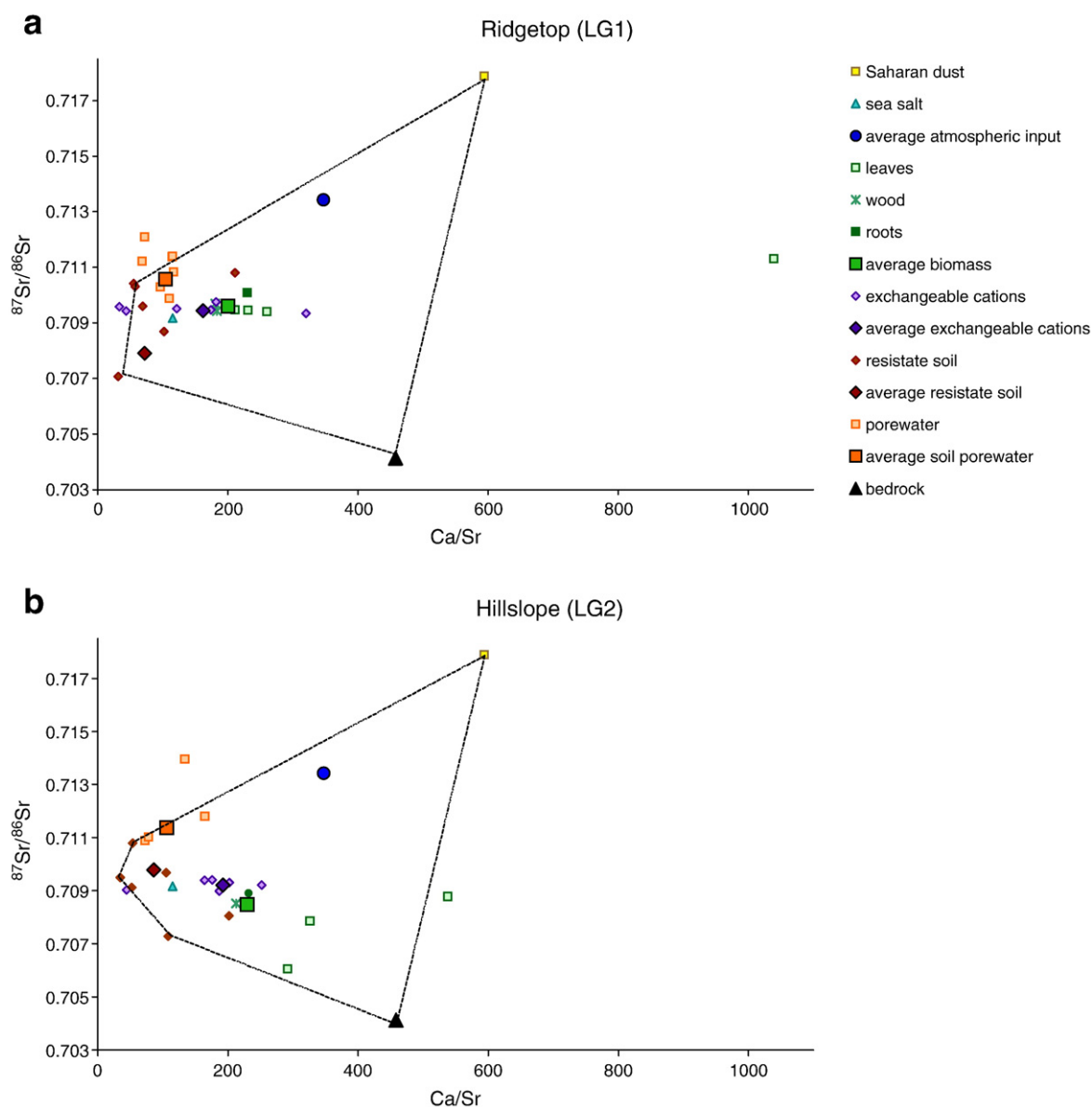


Fig. 7. Ca/Sr plotted against $^{87}\text{Sr}/^{86}\text{Sr}$ ratios of samples at LG1 (a) and LG2 (b). Dashed lines delineate a mixing area defined by possible input end-members (atmospheric, bedrock, and resistate soil) to biomass, exchangeable cations, and porewaters. These polygons envelop areas that include all the Ca/Sr and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios that could derive from simple mixing of two or more potential end-members. Ca/Sr fractionation is required to explain samples plotting outside these areas.

ratios in porewater relative to throughfall indicates some weathering inputs from soil minerals occurs. A small weathering flux from a radiogenic source such as altered biotite in soil, which is expected to have a very low Ca/Sr ratio, may account for the observed porewater values. Another possible mechanism for the shift in soil porewater to lower Ca/Sr ratios is preferential removal of Ca onto soil organic matter. Greater affinity of Ca than Sr for organic matter has been observed in previous work (Baes and Bloom, 1988). Other authors have argued that Sr has a slightly stronger affinity for ion exchange sites than Ca (Bruggenwert and Kamphorst, 1982; Sposito, 1989). It is likely that affinities for ion exchange reactions for specific alkaline earth cations are dependent on a variety of soil factors including mineralogy, organic matter content and pH, and therefore will be variable between sites. For example, Drouet and Herbauts (2008) studying soils in Belgium found either equal affinity for Ca and Sr on the soil exchange complex, or a preference for Ca over Sr in the mineral soil and a preference for Sr over Ca in the humic horizons. In contrast, Poszwa et al. (2000), studying acidic soils in Switzerland and France, found a preference for Ca over Sr exchanged on soil organic

matter. Another mechanism that could explain lower Ca/Sr ratios in soil porewater as compared to throughfall is fractionation at the point of root uptake, with preferential uptake of Ca over Sr. Unfortunately our data are not sufficient to resolve the relative importance of these two potential fractionation mechanisms in this system.

While the literature shows that dominant Ca–Sr fractionation mechanisms and degrees of fractionation vary between sites based on soil types and individual species (Poszwa et al., 2000; Dasch et al., 2006), some general patterns appear to be consistent within the studies that have been done in the tropics. The Ca/Sr ratios of ecosystem compartments in the Rio Icacos watershed (Table 5) are similar to those in a neighboring watershed. Jordan et al. (1972) compiled data on Ca and Sr concentrations from the nearby El Verde research site, and found Ca/Sr molar ratios in precipitation, leaves, wood, and soil of, respectively, 340, 309, 199, and 43. This suggests that the pattern of higher Ca/Sr ratios in leaves as compared to wood and low Ca/Sr ratios in soils is common in the area.

The limited data available from other studies analyzing plant and soil Ca/Sr data in the tropics also shows a similar pattern to that seen

in the Icacos watershed, with high Ca/Sr ratios in biomass and lower Ca/Sr ratios in soils. A study in French Guyana found that the mean foliar Ca/Sr ratio was 160, exchangeable cation Ca/Sr ratios ranged between 169 and 288, and bulk soil Ca/Sr ratios varied between 1 and 8 (Poszwa et al., 2002). In Hawaiian soils that are 150 ka and older, the foliar Ca/Sr ratios range from 160 to 273, exchangeable cation Ca/Sr ratios range between 91 and 182 while the bulk soil Ca/Sr ratios range between 4 and 17 (Wiegand et al., 2005). Younger Hawaiian soils (<150 ka), however, do not show clear differences in the Ca/Sr ratios of these reservoirs, suggesting that this pattern may only develop over time in more highly weathered soils.

6. Conclusions

Ca/Sr and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios of plant and soil samples from the Rio Icacos basin illustrate large contributions from bedrock weathering as well as heterogeneity in nutrient sources on small spatial scales. Using bedrock as the weathering end-member, on a steep hillslope with 3.6 m of regolith, the vegetation derives 67% of its Sr from weathering, while on a ridgetop site with 8.5 m of regolith, the vegetation derives 42% of its Sr from weathering. These results show the benefit of isotopic studies in determining nutrient sources that would have been missed otherwise. Simply measuring Ca contents of ecosystem reservoirs and the atmospheric Ca flux, together with observation of shallow rooting depths and deep highly weathered regolith would have led us to believe this system would be much more atmospherically dominated. Our results suggest that soil sampling could easily underestimate the nutrient pools utilized by plants, and thus misrepresent the importance of Ca, Mg, and K availability to ecosystem productivity. We suggest that the presence of bedrock corestones in the soil may account for the bedrock weathering contribution to nutrient sources in the Rio Icacos watershed. The Sr isotopic data are also particularly useful in highlighting dissimilarities between the $^{87}\text{Sr}/^{86}\text{Sr}$ ratios of porewater and the soil exchangeable cations pool. This signifies that inefficient exchange occurs between these two reservoirs, which is likely the result of preferential flow in macropores.

Vegetation in the Icacos basin will respond on a timescale of $\sim 10^3$ to a shift in the source of base cations because of the small size of the actively cycled pool relative to the magnitude of atmospheric inputs. Evidence of bedrock-derived nutrients presently seen in Icacos vegetation indicates that weathering is an ongoing important source of nutrients and not an artifact of efficient internal recycling.

Ca/Sr ratios reveal that fractionation between these two elements occurs during biogeochemical cycling in the Icacos basin. The pattern of declining Ca/Sr ratios, with Ca/Sr ratios of 350 in leaves, 162 in exchangeable cations, 104 in soil porewater, and 72 in resistate soil at the ridgetop site is consistent with patterns seen elsewhere in highly weathered tropical systems. High Ca/Sr ratios in leaves has also been observed in temperate systems, and ascribed to fractionation during cation transport within plants. The cause of low Ca/Sr ratios in older tropical soils is not well understood.

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